Cyber Threat Detection System

Upgraded Project: Real-Time IDS with Hybrid ML + Deep Learning + Live Traffic Capture

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Foundations of Data Science | End-of-Semester Project

TECH STACK OVERVIEW

Live Packet Capture: Scapy (Python) | Feature Extraction: CICFlowMeter-style pipeline

Models: Random Forest + XGBoost + LSTM Neural Network (ensemble) | Dataset: CICIDS-2017

Backend: FastAPI | Frontend: React dashboard | Explainability: SHAP values

1. Why This Upgrade Matters

Your original proposal used a single Random Forest classifier on a static CSV file. That is a solid foundation, but not enough to impress at end-of-semester level for a cybersecurity-focused data science project. The upgraded version adds three major dimensions that make it genuinely impressive:

- Ensemble model: Random Forest + XGBoost + LSTM gives significantly better performance and shows you understand different model families (tree-based, gradient boosting, sequence models).
- Live traffic capture: Using Scapy to sniff real packets and convert them to flow features means the system actually runs in real time, not just on a CSV file. This is the single biggest upgrade.
- Explainability layer (SHAP): For cybersecurity, being able to say WHY the model flagged something is critical. This is an advanced, industry-relevant feature that will stand out.

ABOUT PACKET TRACER (your friend's suggestion)

Cisco Packet Tracer is a network simulation tool used in networking courses. It does NOT produce real network traffic that Python can capture. Do NOT use it for this project. Instead, use Scapy (a Python library) to capture real packets from your machine's network interface. This is the professional approach and is far more impressive. Scapy works on Windows, Mac, and Linux.

2. Complete File & Folder Structure

Create this exact structure in your project folder. Each file is explained in later sections.

```
cyber-threat-detection/
  README.md
  requirements.txt
  .env                          # API keys if needed

  data/
    raw/                        # Place CICIDS-2017 CSVs here
    processed/
      cleaned.csv
      features.csv
    models/
      rf_model.pkl              # Trained Random Forest
      xgb_model.pkl             # Trained XGBoost
      lstm_model.h5             # Trained LSTM
      scaler.pkl                # StandardScaler
      label_encoder.pkl         # LabelEncoder
      shap_explainer.pkl        # SHAP explainer
```

```
src/
  preprocessing/
    __init__.py
    clean.py                # Data cleaning pipeline
    feature_engineer.py     # Feature selection & engineering
  models/
    __init__.py
    train_rf.py             # Random Forest training
    train_xgb.py            # XGBoost training
    train_lstm.py           # LSTM training
    ensemble.py             # Combines all three models
    evaluate.py             # Metrics, confusion matrix
    explain.py              # SHAP explainability
  capture/
    __init__.py
    live_capture.py         # Scapy packet sniffer
    flow_builder.py         # Converts packets -> flow features
  api/
    __init__.py
    main.py                 # FastAPI app
    schemas.py              # Pydantic request/response models
    predictor.py            # Loads models, runs inference
```

```
frontend/
  index.html
  app.js                    # React dashboard
  style.css

notebooks/
  01_eda.ipynb              # Exploratory Data Analysis
  02_training.ipynb         # Model training walkthrough
  03_evaluation.ipynb       # Results and comparisons

scripts/
  train_all.sh              # Run all training in sequence
  start_api.sh              # Launch FastAPI server
  start_capture.sh          # Start live packet capture
```

3. Step-by-Step: What To Do

Step 1: Download the Dataset

Go to https://www.unb.ca/cic/datasets/ids-2017.html and download the CICIDS-2017 dataset. It is free for academic use. You will get several CSV files (Monday, Tuesday, Wednesday, etc.). Place all CSVs inside data/raw/.

Step 2: Install Dependencies

Create a requirements.txt with these exact contents:

```
pandas>=1.5.0

numpy>=1.23.0

scikit-learn>=1.2.0

xgboost>=1.7.0

tensorflow>=2.11.0

keras>=2.11.0

shap>=0.41.0

scapy>=2.5.0

fastapi>=0.95.0

uvicorn>=0.21.0

pydantic>=1.10.0

imbalanced-learn>=0.10.0

matplotlib>=3.6.0

seaborn>=0.12.0

joblib>=1.2.0

python-multipart>=0.0.6
```

Then run:

```
pip install -r requirements.txt
```

WINDOWS NOTE: Scapy on Windows requires Npcap. Download and install it from https://npcap.com before running any live capture code. Run your terminal as Administrator when using Scapy. On Mac/Linux no extra steps needed.

4. Full Source Code

4.1 src/preprocessing/clean.py

This script loads all CICIDS CSVs, combines them, removes bad rows, and saves a clean file.

```
import pandas as pd
import numpy as np
import glob, os

RAW_DIR  = 'data/raw'
OUT_PATH = 'data/processed/cleaned.csv'

def load_all_csvs():
    files = glob.glob(os.path.join(RAW_DIR, '*.csv'))
    print(f'Found {len(files)} CSV files')
    frames = []
    for f in files:
        df = pd.read_csv(f, encoding='latin-1')
        df.columns = df.columns.str.strip()  # remove whitespace from headers
        frames.append(df)
    return pd.concat(frames, ignore_index=True)

def clean(df):
    print(f'Raw shape: {df.shape}')
    # Drop rows with infinity or NaN
    df.replace([np.inf, -np.inf], np.nan, inplace=True)
    df.dropna(inplace=True)
    # Drop constant columns
```

```
    df = df.loc[:, df.nunique() > 1]
    # Standardize label column
    label_col = ' Label' if ' Label' in df.columns else 'Label'
    df.rename(columns={label_col: 'Label'}, inplace=True)
    df['Label'] = df['Label'].str.strip()
    print(f'Clean shape: {df.shape}')
    print(df['Label'].value_counts())
    return df


if __name__ == '__main__':
    df = load_all_csvs()
    df = clean(df)
    os.makedirs('data/processed', exist_ok=True)
    df.to_csv(OUT_PATH, index=False)
    print(f'Saved to {OUT_PATH}')
```

4.2 src/preprocessing/feature_engineer.py

Selects the most useful features, handles class imbalance with SMOTE, and prepares train/test splits.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.feature_selection import SelectKBest, f_classif
from imblearn.over_sampling import SMOTE
import joblib, os


CLEAN_PATH = 'data/processed/cleaned.csv'
```

```
FEAT_PATH  = 'data/processed/features.csv'
N_FEATURES = 30  # top features to keep

def prepare():
    df = pd.read_csv(CLEAN_PATH)

    le = LabelEncoder()
    df['Label_enc'] = le.fit_transform(df['Label'])
    joblib.dump(le, 'data/models/label_encoder.pkl')

    X = df.drop(columns=['Label', 'Label_enc'])
    y = df['Label_enc']

    # Select top N features
    selector = SelectKBest(f_classif, k=N_FEATURES)
    X_sel = selector.fit_transform(X, y)
    selected_cols = X.columns[selector.get_support()].tolist()
    print('Selected features:', selected_cols)

    # Scale
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X_sel)
    joblib.dump(scaler, 'data/models/scaler.pkl')
    joblib.dump(selected_cols, 'data/models/feature_cols.pkl')

    # Split
```

```
    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled, y, test_size=0.2, stratify=y, random_state=42)

    # SMOTE to fix class imbalance
    sm = SMOTE(random_state=42)
    X_train, y_train = sm.fit_resample(X_train, y_train)
    print(f'After SMOTE: {X_train.shape}')

    return X_train, X_test, y_train, y_test, le

if __name__ == '__main__':
    os.makedirs('data/models', exist_ok=True)
    prepare()
```

4.3 src/models/train_rf.py

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report
import joblib
from src.preprocessing.feature_engineer import prepare

def train():
    X_train, X_test, y_train, y_test, le = prepare()
    print('Training Random Forest...')
    clf = RandomForestClassifier(
        n_estimators=200,
        max_depth=None,
        min_samples_split=5,
```

```
        class_weight='balanced',
        n_jobs=-1,
        random_state=42
    )
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    print(classification_report(y_test, y_pred, target_names=le.classes_))
    joblib.dump(clf, 'data/models/rf_model.pkl')
    print('Saved: data/models/rf_model.pkl')

if __name__ == '__main__':
    train()
```

4.4 src/models/train_xgb.py

```
from xgboost import XGBClassifier
from sklearn.metrics import classification_report
import joblib
from src.preprocessing.feature_engineer import prepare

def train():
    X_train, X_test, y_train, y_test, le = prepare()
    print('Training XGBoost...')
    clf = XGBClassifier(
        n_estimators=300,
        max_depth=6,
        learning_rate=0.1,
        subsample=0.8,
```

```
        colsample_bytree=0.8,
        use_label_encoder=False,
        eval_metric='mlogloss',
        n_jobs=-1,
        random_state=42
    )
    clf.fit(X_train, y_train,
            eval_set=[(X_test, y_test)],
            verbose=50)
    y_pred = clf.predict(X_test)
    print(classification_report(y_test, y_pred, target_names=le.classes_))
    joblib.dump(clf, 'data/models/xgb_model.pkl')
    print('Saved: data/models/xgb_model.pkl')


if __name__ == '__main__':
    train()
```

4.5 src/models/train_lstm.py

LSTM (Long Short-Term Memory) is a type of neural network that can learn patterns over sequences. Network flows have temporal order, making LSTM a natural fit. This is what will impress your professor most.

```
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
from tensorflow.keras.utils import to_categorical
import joblib
from src.preprocessing.feature_engineer import prepare
```

```
def train():
    X_train, X_test, y_train, y_test, le = prepare()
    n_classes = len(le.classes_)

    # Reshape for LSTM: (samples, timesteps=1, features)
    X_tr_l = X_train.reshape(X_train.shape[0], 1, X_train.shape[1])
    X_te_l = X_test.reshape(X_test.shape[0], 1, X_test.shape[1])
    y_tr_c = to_categorical(y_train, n_classes)
    y_te_c = to_categorical(y_test, n_classes)

    model = Sequential([
        LSTM(128, input_shape=(1, X_train.shape[1]),
             return_sequences=True),
        BatchNormalization(),
        Dropout(0.3),
        LSTM(64),
        BatchNormalization(),
        Dropout(0.3),
        Dense(64, activation='relu'),
        Dropout(0.2),
        Dense(n_classes, activation='softmax')
    ])

    model.compile(optimizer='adam',
                  loss='categorical_crossentropy',
                  metrics=['accuracy'])
```

```
    model.summary()


    callbacks = [
        EarlyStopping(patience=5, restore_best_weights=True),
        ModelCheckpoint('data/models/lstm_model.h5', save_best_only=True)
    ]


    model.fit(X_tr_l, y_tr_c,
              validation_data=(X_te_l, y_te_c),
              epochs=30,
              batch_size=1024,
              callbacks=callbacks)


if __name__ == '__main__':
    train()
```

4.6 src/models/ensemble.py

The ensemble combines all three model predictions using soft voting (averaging probabilities). This is the key to beating any single model.

```
import numpy as np
import joblib
from tensorflow.keras.models import load_model


class EnsemblePredictor:
    def __init__(self):
        self.rf  = joblib.load('data/models/rf_model.pkl')
        self.xgb = joblib.load('data/models/xgb_model.pkl')
        self.lstm = load_model('data/models/lstm_model.h5')
```

```
        self.scaler = joblib.load('data/models/scaler.pkl')
        self.le     = joblib.load('data/models/label_encoder.pkl')
        self.feature_cols = joblib.load('data/models/feature_cols.pkl')

    def predict(self, X_scaled: np.ndarray):
        """
        X_scaled: 2D numpy array, shape (n_samples, n_features)
        Returns: list of dicts with label, confidence, risk_level
        """
        p_rf  = self.rf.predict_proba(X_scaled)
        p_xgb = self.xgb.predict_proba(X_scaled)

        X_lstm = X_scaled.reshape(X_scaled.shape[0], 1, X_scaled.shape[1])
        p_lstm = self.lstm.predict(X_lstm)

        # Soft vote: average probabilities with weights
        p_avg = (0.35 * p_rf + 0.35 * p_xgb + 0.30 * p_lstm)

        predicted_idx = np.argmax(p_avg, axis=1)
        confidence    = np.max(p_avg, axis=1)
        labels        = self.le.inverse_transform(predicted_idx)

        results = []
        for label, conf in zip(labels, confidence):
            results.append({
                'label': label,
```

```
                'confidence': round(float(conf) * 100, 2),
                'risk_level': _risk(label),
            })
        return results


def _risk(label):
    label = label.upper()
    if 'BENIGN' in label:    return 'NONE'
    if 'DDOS' in label:      return 'CRITICAL'
    if 'BOT' in label:       return 'HIGH'
    if 'BRUTE' in label:     return 'HIGH'
    if 'INFILTR' in label:   return 'HIGH'
    if 'PORT' in label:      return 'MEDIUM'
    if 'SQL' in label:       return 'MEDIUM'
    if 'XSS' in label:       return 'MEDIUM'
    if 'WEB' in label:       return 'MEDIUM'
    return 'LOW'
```

4.7 src/models/explain.py (SHAP Explainability)

SHAP values tell you which features pushed the model toward a particular decision. This is what makes the system production-grade and is an excellent talking point in your presentation.

```
import shap

import joblib

import numpy as np

import matplotlib.pyplot as plt



def build_shap_explainer():

    rf = joblib.load('data/models/rf_model.pkl')

    # TreeExplainer is fast for Random Forest / XGBoost

    explainer = shap.TreeExplainer(rf)

    joblib.dump(explainer, 'data/models/shap_explainer.pkl')

    print('SHAP explainer saved.')

    return explainer



def explain_single(X_sample: np.ndarray, feature_names: list):

    """

    X_sample: 1D array of one flow's features

    Returns: dict of {feature: shap_value}

    """

    explainer = joblib.load('data/models/shap_explainer.pkl')

    shap_values = explainer.shap_values(X_sample.reshape(1, -1))

    # Take values for the predicted class

    pred_class = np.argmax(explainer.expected_value)

    sv = shap_values[pred_class][0]

    return dict(sorted(zip(feature_names, sv),
```

```
                        key=lambda x: abs(x[1]), reverse=True)[:10])


def plot_summary(X_test, feature_names):
    explainer = joblib.load('data/models/shap_explainer.pkl')
    shap_values = explainer.shap_values(X_test[:500])
    shap.summary_plot(shap_values[0], X_test[:500],
                      feature_names=feature_names, show=True)


if __name__ == '__main__':
    build_shap_explainer()
```

4.8 src/models/evaluate.py

```
import joblib, numpy as np, matplotlib.pyplot as plt, seaborn as sns
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_auc_score
)
from src.preprocessing.feature_engineer import prepare
from src.models.ensemble import EnsemblePredictor


def full_evaluation():
    _, X_test, _, y_test, le = prepare()
    predictor = EnsemblePredictor()
    results = predictor.predict(X_test)
    y_pred_labels = [r['label'] for r in results]
    y_pred_enc = le.transform(y_pred_labels)

    print('=== ENSEMBLE CLASSIFICATION REPORT ===')
```

```
    print(classification_report(y_test, y_pred_enc, target_names=le.classes_))


    cm = confusion_matrix(y_test, y_pred_enc)
    plt.figure(figsize=(12, 10))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=le.classes_, yticklabels=le.classes_)
    plt.title('Ensemble Confusion Matrix')
    plt.tight_layout()
    plt.savefig('confusion_matrix.png', dpi=150)
    print('Confusion matrix saved.')


if __name__ == '__main__':
    full_evaluation()
```

4.9 src/capture/flow_builder.py (Real-Time Core)

This is the module that converts raw packets into the same 78-feature format that CICIDS-2017 uses. This is the hardest but most impressive part of the whole project.

```
import time

from collections import defaultdict

import numpy as np



class FlowRecord:

    """Tracks a single TCP/UDP network flow."""

    def __init__(self, key):

        self.key = key

        self.start_time = time.time()

        self.last_time  = time.time()

        self.fwd_pkts, self.bwd_pkts = [], []

        self.fwd_bytes, self.bwd_bytes = 0, 0

        self.fwd_iat, self.bwd_iat = [], []

        self.flags = defaultdict(int)  # SYN, ACK, FIN, RST, PSH, URG



    def add_packet(self, pkt_len, direction, tcp_flags=None, timestamp=None):

        now = timestamp or time.time()

        iat = now - self.last_time

        self.last_time = now

        if direction == 'fwd':

            self.fwd_pkts.append(pkt_len)

            self.fwd_bytes += pkt_len

            self.fwd_iat.append(iat)

        else:
```

```
            self.bwd_pkts.append(pkt_len)
            self.bwd_bytes += pkt_len
            self.bwd_iat.append(iat)
        if tcp_flags:
            for f in ['S', 'A', 'F', 'R', 'P', 'U']:
                if f in str(tcp_flags):
                    self.flags[f] += 1

    def to_feature_vector(self) -> dict:
        duration = max(self.last_time - self.start_time, 1e-6)
        total_pkts  = len(self.fwd_pkts) + len(self.bwd_pkts)
        all_iat = self.fwd_iat + self.bwd_iat

        def stats(lst):
            if not lst: return 0, 0, 0, 0
            a = np.array(lst)
            return float(a.mean()), float(a.std()), float(a.min()), float(a.max())

        fwd_m, fwd_s, fwd_mn, fwd_mx = stats(self.fwd_pkts)
        bwd_m, bwd_s, bwd_mn, bwd_mx = stats(self.bwd_pkts)
        iat_m, iat_s, iat_mn, iat_mx = stats(all_iat)

        return {
            'Flow Duration':        duration * 1e6,
            'Total Fwd Packets':    len(self.fwd_pkts),
            'Total Bwd Packets':    len(self.bwd_pkts),
```

```
            'Total Length Fwd':     self.fwd_bytes,
            'Total Length Bwd':     self.bwd_bytes,
            'Fwd Packet Length Mean': fwd_m,
            'Fwd Packet Length Std':  fwd_s,
            'Fwd Packet Length Min':  fwd_mn,
            'Fwd Packet Length Max':  fwd_mx,
            'Bwd Packet Length Mean': bwd_m,
            'Bwd Packet Length Std':  bwd_s,
            'Bwd Packet Length Min':  bwd_mn,
            'Bwd Packet Length Max':  bwd_mx,
            'Flow Bytes/s':   self.fwd_bytes + self.bwd_bytes / duration,
            'Flow Packets/s': total_pkts / duration,
            'Flow IAT Mean':  iat_m,
            'Flow IAT Std':   iat_s,
            'Flow IAT Min':   iat_mn,
            'Flow IAT Max':   iat_mx,
            'Fwd IAT Mean':   stats(self.fwd_iat)[0],
            'Bwd IAT Mean':   stats(self.bwd_iat)[0],
            'SYN Flag Count': self.flags['S'],
            'ACK Flag Count': self.flags['A'],
            'FIN Flag Count': self.flags['F'],
            'RST Flag Count': self.flags['R'],
            'PSH Flag Count': self.flags['P'],
            'URG Flag Count': self.flags['U'],
            'Avg Fwd Segment Size': fwd_m,
            'Avg Bwd Segment Size': bwd_m,
            'Subflow Fwd Packets':  len(self.fwd_pkts),
```

```
            'Subflow Bwd Packets':  len(self.bwd_pkts),
        }


class FlowTracker:
    """Maintains all active flows, expires old ones."""
    FLOW_TIMEOUT = 120  # seconds

    def __init__(self):
        self.flows = {}
        self.completed = []

    def _key(self, src_ip, dst_ip, sport, dport, proto):
        return (min(src_ip, dst_ip), max(src_ip, dst_ip),
                min(sport, dport), max(sport, dport), proto)

    def process_packet(self, src_ip, dst_ip, sport, dport,
                       proto, pkt_len, tcp_flags=None):
        key = self._key(src_ip, dst_ip, sport, dport, proto)
        direction = 'fwd' if (src_ip, sport) <= (dst_ip, dport) else 'bwd'

        if key not in self.flows:
            self.flows[key] = FlowRecord(key)

        self.flows[key].add_packet(pkt_len, direction, tcp_flags)
```

```
    # Expire old flows
    now = time.time()
    expired = [k for k, v in self.flows.items()
               if now - v.last_time > self.FLOW_TIMEOUT]
    for k in expired:
        self.completed.append(self.flows.pop(k))


def get_completed_features(self):
    out = [f.to_feature_vector() for f in self.completed]
    self.completed.clear()
    return out
```

4.10 src/capture/live_capture.py (Scapy Sniffer)

This is what makes the system real-time. Scapy captures packets from your network card, passes them to FlowTracker, builds features, and sends them to the ensemble for classification.

```
from scapy.all import sniff, IP, TCP, UDP

import numpy as np, joblib, threading, queue, time

from src.capture.flow_builder import FlowTracker

from src.models.ensemble import EnsemblePredictor



result_queue = queue.Queue()  # thread-safe output



def start_capture(interface=None, duration=None):

    """

    interface: e.g. 'eth0' on Linux, None for default

    duration: seconds to run (None = forever)

    """

    tracker   = FlowTracker()

    predictor = EnsemblePredictor()

    scaler    = joblib.load('data/models/scaler.pkl')

    feat_cols = joblib.load('data/models/feature_cols.pkl')



    def packet_callback(pkt):

        if IP not in pkt: return

        src  = pkt[IP].src

        dst  = pkt[IP].dst

        plen = len(pkt)

        proto = pkt[IP].proto
```

```
            sport, dport, flags = 0, 0, None
            if TCP in pkt:
                sport = pkt[TCP].sport
                dport = pkt[TCP].dport
                flags = pkt[TCP].flags
            elif UDP in pkt:
                sport = pkt[UDP].sport
                dport = pkt[UDP].dport

            tracker.process_packet(src, dst, sport, dport, proto, plen, flags)

            # Check completed flows every packet
            features_list = tracker.get_completed_features()
            for feat_dict in features_list:
                _classify_flow(feat_dict, feat_cols, scaler, predictor)

    print(f'[*] Starting live capture on {interface or "default interface"}...')
    sniff(iface=interface, prn=packet_callback,
          store=False, timeout=duration)

def _classify_flow(feat_dict, feat_cols, scaler, predictor):
    row = np.array([feat_dict.get(c, 0.0) for c in feat_cols])
    row_scaled = scaler.transform(row.reshape(1, -1))
    result = predictor.predict(row_scaled)[0]
    result['flow_key'] = str(list(feat_dict.values())[:3])
    result['timestamp'] = time.time()
```

```
    result_queue.put(result)
    if result['risk_level'] != 'NONE':
        print(f"[ALERT] {result['label']}  |  Conf: {result['confidence']}%"
              f"  |  Risk: {result['risk_level']}")


if __name__ == '__main__':
    start_capture(duration=60)  # capture for 60 seconds
```

4.11 src/api/main.py (FastAPI Backend)

```
from fastapi import FastAPI, HTTPException

from fastapi.middleware.cors import CORSMiddleware

from src.api.schemas import FlowInput, PredictionOut

from src.api.predictor import Predictor

from src.capture.live_capture import result_queue

import threading, asyncio


app = FastAPI(title='Cyber Threat Detection API')


app.add_middleware(CORSMiddleware,

    allow_origins=['*'], allow_methods=['*'], allow_headers=['*'])


predictor = Predictor()


@app.post('/predict', response_model=PredictionOut)

async def predict(flow: FlowInput):

    """Accept a single flow's features and return a classification."""

    try:

        result = predictor.predict_one(flow.dict())

        return result

    except Exception as e:

        raise HTTPException(status_code=500, detail=str(e))


@app.get('/live-alerts')

async def live_alerts():
```

```
    """Return all pending real-time alerts from the capture queue."""
    alerts = []
    while not result_queue.empty():
        alerts.append(result_queue.get_nowait())
    return {'alerts': alerts, 'count': len(alerts)}


@app.get('/health')
def health():
    return {'status': 'ok', 'models': ['rf', 'xgb', 'lstm']}


# To run: uvicorn src.api.main:app --reload --port 8000
```

4.12 src/api/schemas.py

```
from pydantic import BaseModel
from typing import Optional


class FlowInput(BaseModel):
    Flow_Duration: float
    Total_Fwd_Packets: int
    Total_Bwd_Packets: int
    Total_Length_Fwd: float
    Total_Length_Bwd: float
    Fwd_Packet_Length_Mean: float
    Bwd_Packet_Length_Mean: float
    Flow_Bytes_s: float
    Flow_Packets_s: float
    Flow_IAT_Mean: float
```

```
    SYN_Flag_Count: int
    ACK_Flag_Count: int
    FIN_Flag_Count: int
    RST_Flag_Count: int
    # Add remaining features as needed


class PredictionOut(BaseModel):
    label: str
    confidence: float
    risk_level: str
    top_features: Optional[dict] = None  # SHAP values
```

4.13 src/api/predictor.py

```
import numpy as np, joblib
from src.models.ensemble import EnsemblePredictor
from src.models.explain import explain_single


class Predictor:
    def __init__(self):
        self.ensemble  = EnsemblePredictor()
        self.scaler    = joblib.load('data/models/scaler.pkl')
        self.feat_cols = joblib.load('data/models/feature_cols.pkl')


    def predict_one(self, flow_dict: dict) -> dict:
        row = np.array([flow_dict.get(c, 0.0) for c in self.feat_cols])
        row_scaled = self.scaler.transform(row.reshape(1, -1))
        result = self.ensemble.predict(row_scaled)[0]
```

```
# Attach SHAP explanation
try:
    result['top_features'] = explain_single(
        row_scaled[0], self.feat_cols)
except Exception:
    result['top_features'] = {}
return result
```

5. Frontend Dashboard (frontend/app.js)

A React-based dashboard that polls the API for live alerts and visualizes them. Copy the HTML below into frontend/index.html.

```
<!DOCTYPE html>

<html>

<head>

  <title>Cyber Threat Dashboard</title>

  <script src='https://unpkg.com/react@18/umd/react.development.js'></script>

  <script src='https://unpkg.com/react-dom@18/umd/react-dom.development.js'></script>

  <script src='https://unpkg.com/babel-standalone@6/babel.min.js'></script>

  <link rel='stylesheet' href='style.css'>

</head>

<body>

  <div id='root'></div>

  <script type='text/babel' src='app.js'></script>

</body>

</html>
```

app.js - simplified key logic:

```
const API = 'http://localhost:8000';



function Dashboard() {
```

```
const [alerts, setAlerts] = React.useState([]);
const [stats, setStats] = React.useState({total:0, critical:0, high:0});

React.useEffect(() => {
  const interval = setInterval(async () => {
    const res = await fetch(`${API}/live-alerts`);
    const data = await res.json();
    if (data.alerts.length > 0) {
      setAlerts(prev => [...data.alerts, ...prev].slice(0, 100));
      setStats(prev => ({
        total: prev.total + data.count,
        critical: prev.critical + data.alerts.filter(
          a => a.risk_level === 'CRITICAL').length,
        high: prev.high + data.alerts.filter(
          a => a.risk_level === 'HIGH').length
      }));
    }
  }, 2000);  // poll every 2 seconds
  return () => clearInterval(interval);
}, []);
```

```
  const riskColor = (r) => ({
    CRITICAL: '#DC2626', HIGH: '#D97706',
    MEDIUM: '#2563EB', LOW: '#16A34A', NONE: '#6B7280'
  }[r] || '#6B7280');

  return (
    <div style={{fontFamily:'Arial', padding:'20px', background:'#0F172A',
minHeight:'100vh'}}>
      <h1 style={{color:'#60A5FA'}}>Cyber Threat Detection - Live Dashboard</h1>
      <div style={{display:'flex', gap:'20px', marginBottom:'24px'}}>
        {[['Total Flows', stats.total, '#60A5FA'],
          ['Critical', stats.critical, '#DC2626'],
          ['High', stats.high, '#D97706']].map(([l,v,c]) => (
          <div style={{background:'#1E293B', padding:'16px', borderRadius:'8px',
                       borderLeft:`4px solid ${c}`, minWidth:'140px'}}>
            <div style={{color:'#94A3B8', fontSize:'12px'}}>{l}</div>
            <div style={{color:c, fontSize:'28px', fontWeight:'bold'}}>{v}</div>
          </div>
        ))}
      </div>
      <table style={{width:'100%', borderCollapse:'collapse', color:'#E2E8F0'}}>
        <thead>
```

```
            <tr style={{background:'#1E293B'}}>
              {['Time','Label','Confidence','Risk'].map(h => (
                <th style={{padding:'10px', textAlign:'left'}}>{h}</th>
              ))}
            </tr>
          </thead>
          <tbody>
            {alerts.map((a, i) => (
              <tr key={i} style={{borderBottom:'1px solid #1E293B'}}>
                <td style={{padding:'8px'}}>{new
Date(a.timestamp*1000).toLocaleTimeString()}</td>
                <td style={{padding:'8px'}}>{a.label}</td>
                <td style={{padding:'8px'}}>{a.confidence}%</td>
                <td style={{padding:'8px'}}>
                  <span style={{background:riskColor(a.risk_level),
                                padding:'2px 8px', borderRadius:'4px', fontSize:'12px'}}>
                    {a.risk_level}</span>
                </td>
              </tr>
            ))}
          </tbody>
        </table>
```

```
    </div>
  );
}


ReactDOM.createRoot(document.getElementById('root')).render(<Dashboard/>);
```

6. How To Run Everything

6.1 Training All Models

Run these in order from the project root. Training will take 20-60 minutes total depending on your machine.

```
# Step 1: Clean the data

python -m src.preprocessing.clean



# Step 2: Train Random Forest

python -m src.models.train_rf



# Step 3: Train XGBoost

python -m src.models.train_xgb



# Step 4: Train LSTM (slowest - can use GPU if available)

python -m src.models.train_lstm



# Step 5: Build SHAP explainer

python -m src.models.explain



# Step 6: Evaluate ensemble

python -m src.models.evaluate
```

6.2 Starting the API Server

```
uvicorn src.api.main:app --reload --host 0.0.0.0 --port 8000



# API docs available at: http://localhost:8000/docs

# Test with curl:
```

```
curl http://localhost:8000/health
```

6.3 Starting Live Capture (run as admin/root)

```
# On Linux/Mac:

sudo python -m src.capture.live_capture



# On Windows (run terminal as Administrator):

python -m src.capture.live_capture



# Capture for 30 seconds (for demo/testing):

python -c "from src.capture.live_capture import start_capture;
start_capture(duration=30)"
```

6.4 Opening the Dashboard

```
# Open frontend/index.html in your browser.

# Or use a simple local server:

cd frontend

python -m http.server 3000

# Then open: http://localhost:3000
```

7. Model Performance Benchmarks (Expected)

Based on published research using CICIDS-2017, you should expect approximately these results after training your ensemble. Individual model results will be lower.

Model	Expected Accuracy
Random Forest (baseline)	~97-98%
XGBoost	~97-99%
LSTM	~95-97%
Ensemble (all three)	~98-99.5%

KEY METRIC TO HIGHLIGHT TO YOUR PROFESSOR: Precision and Recall per class.

Overall accuracy is misleading because BENIGN traffic dominates the dataset (~80%).

Show per-class F1 scores. A model that just predicts 'BENIGN' always gets 80% accuracy

but is useless. Your ensemble should show high recall (>90%) on all attack classes.

8. What To Put In Your Presentation

What Makes This Impressive vs. the Original

Aspect	Your Upgraded Version
Models used	RF + XGBoost + LSTM ensemble (3 model families)
Input data	CICIDS-2017 + live Scapy packet capture
Class imbalance	Handled with SMOTE oversampling
Feature selection	SelectKBest (top 30 features)
Explainability	SHAP values per prediction
Deployment	FastAPI REST endpoint + React dashboard
Real-time detection	Live packet sniffer with flow builder

Key Points To Make In The Presentation

- Justify why three models: RF handles non-linearity well, XGBoost reduces bias via boosting, LSTM captures any temporal patterns in packet sequences. Ensemble beats any single model.
- Justify SMOTE: Without it, the model over-predicts BENIGN because it dominates the dataset. Show class distribution before and after.
- Justify SHAP: In real security systems, analysts need to know WHY something was flagged, not just that it was. This is the difference between a research toy and a deployable tool.
- Show a live demo: Start the API, start capture, open the dashboard, browse a website, trigger a ping flood or port scan (on your own machine) with nmap, and show the real-time alert appear.
- Packet Tracer vs Scapy: If asked, explain that Packet Tracer is a Cisco simulation tool for learning networking protocols. Scapy captures real packets from a live network interface. The two are completely different. Scapy is the correct tool here.

How To Generate Demo Traffic (Safe, On Your Own Machine)

You can generate attack-like traffic on your own laptop safely for demonstration purposes. These affect only your own machine:

```
# Install nmap first: https://nmap.org



# Simulate a port scan (triggers Port Scan detection):

nmap -sS localhost
```

```
# Simulate SYN flood (triggers DDoS detection) - ONLY on your own machine:
# On Linux:
hping3 -S --flood -V localhost


# Or use Python to generate fast TCP connections to localhost:
python -c "
import socket, threading
def flood():
    for port in range(1, 1000):
        try: socket.create_connection(('127.0.0.1', port), timeout=0.01)
        except: pass
[threading.Thread(target=flood).start() for _ in range(5)]
"
```

9. Common Issues & Fixes

Problem	Solution
Scapy shows 'no such device'	Run as administrator. On Windows, install Npcap from npcap.com
CSV column has leading spaces	clean.py strips headers with .str.strip() - already handled
SMOTE runs out of memory	Reduce dataset: df.sample(500000) before SMOTE
LSTM is very slow	Add GPU: install tensorflow-gpu. Or reduce epochs to 10.
API returns 500 on predict	Check that all .pkl and .h5 files exist in data/models/
Flow features are all zeros	Flow timeout is 120s. Short flows complete after FIN flag.
SHAP is slow	Use shap.TreeExplainer(rf) not DeepExplainer for RF/XGB
Dashboard shows no alerts	Make sure both API (port 8000) and capture are running

10. Grading Checklist

Use this to make sure you have everything before submission:

- Dataset downloaded and cleaned (all CICIDS-2017 CSV files combined)
- Three models trained: Random Forest, XGBoost, LSTM
- Ensemble module combining all three with soft voting
- SHAP explainability working (shap_explainer.pkl exists)
- SMOTE applied to training data to fix class imbalance
- Per-class classification report (not just overall accuracy)
- Confusion matrix generated and saved
- FastAPI backend running and responding at /predict and /live-alerts
- Scapy live capture working (even briefly, for demo)
- React dashboard showing real-time alerts
- Presentation slides explaining: dataset, preprocessing, why each model, ensemble, SHAP, live demo
- EDA notebook showing: class distribution, feature correlations, attack type breakdown
- README.md with setup instructions and project description

Good luck! The real-time capture + ensemble + SHAP combination is genuinely research-grade.

Sanvi Saket & Tejashree | Foundations of Data Science